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# Intercostal nerve cryoablation during surgical stabilization of rib fractures decreases post-operative opioid use, ventilation days, and intensive care days

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## Highlights

- Cryoablation provides lasting nerve blockade and improves outcomes during rib fixation.
- This retrospective study analyzed outcomes during SSRF with and without cryoablation.
- Cryoablated patients required less post-operative total, daily, and discharge MME.
- No increase in pulmonary complications or operative time with cryoablation.

## Abstract

### Background

Intercostal nerve cryoablation is an adjunctive measure that has demonstrated pain control, decrease in opioid consumption, and decrease in hospital length of stay (LOS) in patients who undergo surgical stabilization of rib fractures (SSRF).

## Methods

SSRF patients from January 2015 to September 2021 were retrospectively compared. All patients received multimodal pain regimens post-operatively and the independent variable was intraoperative cryoablation.

## Results

241 patients met inclusion criteria. 51 (21%) underwent intra-operative cryoablation during SSRF and 191 (79%) did not. Patients with standard treatment consumed 9.4 more daily MME ($p$=0.035), consumed 73 percent more post-operative total MME ($p$=0.001), spent 1.55 times as many days in the intensive care unit ($p$=0.013), and spent 3.8 times as many days on the ventilator than patients treated with cryoablation, respectively. Overall hospital LOS, operative case time, pulmonary complications, MME at discharge, and numeric pain scores at discharge were no different (all $p$>0.05).

## Conclusion

Intercostal nerve cryoablation during SSRF is associated with fewer ventilator days, ICU LOS, total post-operative, and daily opioid use without increasing time in the operating room or perioperative pulmonary complications.

## Introduction

Incidence of traumatic rib fractures comprises 10%–20% of blunt injuries and is frequently associated with multisystem trauma [1,2]. Current management strategies rely on pain control, pulmonary hygiene, and early mobility. Even so, rib fractures still comprise a significant cause of morbidity and mortality in those with inadequate pain relief [3,4].

Systemic opioids have remained the pharmacologic workhorse to address pain control in these patients. In the United States, opioid prescriptions have nearly quadrupled without any measured difference in reported pain [5]. In an effort to reduce overprescribing, many institutions have developed a multimodal pain regimen (MMPR) approach to pain control. This approach frequently incorporates both opioid and non-opioid analgesics, topical, and non-pharmacologic therapies. Multimodal analgesia relies on synergistic combinations of medications to decrease opioid dosing requirements [6]. Our institution's MMPR has been well documented in a recent publication [7].

Surgical stabilization of rib fractures (SSRF) provides an effective way to minimize opioid consumption and pulmonary complications by reestablishing physiologic chest wall movement and functionality. Increasingly, patients are benefiting from SSRF to re-establish inspiratory physiology [8]. As technological advancements and refinement of surgical technique continues to accelerate, indications for SSRF have expanded [9,10]. Recent studies have illustrated that SSRF leads to decreased hospital length of stay (LOS), decreased intensive care unit (ICU) LOS, decreased

ventilator days, decreased incidence of pneumonia, and decreased need for opioids [9,11,12]. However, many patients still complain of pain, discomfort, and mild-to-moderate disability up to 1 year following the intervention with current pain management [13,14].

Cryoablation is not a new technology. A useful adjunct in thoracic surgery since the 1970s and the modern Nuss procedure, cryoablation provides lasting regional nerve blockade that spans the immediate post-traumatic and post-operative period [15], [16], [17], [18]. Functionally, application of the cryoablation probe to intercostal nerves of temperatures between -40 to -70°Celsius for 2 min results in Wallerian degeneration and a subsequent analgesic effect that lasts for 6–12 months [11]. Utilizing appropriate technique, risks of neuroma or pseudo-hernia formation are low as the technique does not damage the perineurium nor epineurium, and nearby structures such as vessels and bones are protected [19,20].

Initial case series indicating sustained analgesia are encouraging, however not enough data have accrued for widespread acceptance of the technique [21], [22], [23]. Additionally, no study has evaluated cryoablation's utilization in the context of pre-operative peripheral nerve blocks or opioid infusions. Our institution, as well as others across the nation, have added cryoablation routinely to our multimodal treatment of traumatic rib pain when patients undergo SSRF. The purpose of this study was to analyze our experience with cryoablation in patients undergoing SSRF after traumatic chest wall injury. We hypothesized that cryoablation use during SSRF leads to less opioid consumption, fewer pulmonary complication rates, and decreased hospital LOS than patients who undergo a standard MMPR following SSRF.

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## Methods

After obtaining approval from the Institutional Review Board, a retrospective review of two American College of Surgeons-verified trauma centers within the same system (level 1&3) was performed for all patients that underwent SSRF between January 2015 and September 2021. Patients younger than 18 years of age, those with documented opioid use at home, and those who underwent outpatient SSRF were excluded.

Collected data included age, gender, and comorbidities including previous lung disease …

## Results

Two hundred forty-eight (248) patients underwent SSRF and met criteria in the study period. Five patients were excluded due to home opioid use and two patients excluded due to age making our

study size 241 patients. Of those included, 51 (21%) underwent intraoperative cryoablation during SSRF and 191 (79%) did not. Patient characteristics were similar between the groups with the exception that patients in the standard MMPR group were younger, (standard- median(IQR): 55.0 (42.0–65.0) vs …

## Discussion

This retrospective study of patients who underwent SSRF with and without cryoablation demonstrates that cryoablation reduced MME use and is a safe procedure. Perioperative complications were not increased and no increase in operative time was experienced. This unique study examines in hospital morbidity and the extent of cryoablation effect on opioid demand post-operatively.

Management of pain in trauma patients is a careful balance of relieving traumatic pain without compromising pulmonary …

## Conclusion

Cryoablation provides long-acting regional nerve blockade that outlasts injections and catheter-based delivery systems, providing analgesia throughout the postoperative and post-traumatic period associated with the most severe pain. Local anesthesia in the serratus anterior, paravertebral, or erector spinae plane is frequently utilized by clinicians and has literature to support its use in the traumatic rib fracture patient. In our multivariable analysis, we demonstrate that cryoablation is …

## Funding

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## Declaration of Competing Interest

Dr Bradley W Thomas is an educational consultant for Zimmer Biomedical.

Dr Sean T Dieffenbaugher is a speaker for DePuy Synthes.

No other conflicts of interest or disclosures exist for the remaining authors.

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